Lecture 2

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E1 248
Sliding mode control and its application

Overview

1. Solution of the time-invariant state equation

2. Numerical Example

3. Discrete-time systems

Solution of the time-invariant state equation

Before we solve the vector matrix differential equations, let us review the solution of the scalar differential equation

$$\dot{x} = ax$$

In solving this equation, we may assume a solution $x(t)$ of the form

$$x(t) = b_0 + b_1 t + b_2 t^2 + \cdots + b_k t^k + \cdots$$

Solution of the time-invariant state equation

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$$x(t) = b_0 + b_1 t + b_2 t^2 + \cdots + b_k t^k + \cdots$$

By substituting this assumed solution in the above equation, we obtain

$$b_1 + 2b_2 t + 3b_3 t^2 + \cdots + kb_k t^{k-1} + \cdots = ab_0 + ab_1 t + ab_2 t^2 + \cdots + ab_k t^k + \cdots$$

Equating the coefficient of the equal power of t we obtain

$$
\begin{aligned}
b_1 &= ab_0 \\
b_2 &= \frac{1}{2}ab_1 = \frac{1}{2}a^2 b_0 \\
b_3 &= \frac{1}{3}ab_2 = \frac{1}{6}a^3 b_0 \\
&\vdots \\
b_k &= \frac{1}{k!}a^k b_0
\end{aligned}
$$

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b_k &= \frac{1}{k!}a^k b_0
\end{aligned}
$$

The value of b_0 is determined by substituting $t = 0$, one can get $b_0 = x(0)$. So the solution can be written as

$$
\begin{aligned}
x(t) &= b_0 + ab_o t + \frac{1}{2}a^2 b_0 t^2 + \cdots \\
&= (1 + at + a^2\frac{t^2}{2!} + \cdots)b_0 \\
&= e^{at}b_0 = e^{at}x(0)
\end{aligned}
$$

We shall now solve the vector matrix differential equation

$$\dot{x} = Ax$$

where $x \in \mathbb{R}^n$ and $A \in \mathbb{R}^{n \times n}$

By analogy with the scalar case, we assume that the solution is in the form of a vector power series in t

$$x(t) = b_0 + b_1 t + b_2 t^2 + \cdots + b_k t^k + \cdots$$

If the assumed solution is to be the true solution, then

$$b_1 + 2b_2 t + 3b_3 t^2 + \cdots + k b_k t^{k-1} + \cdots = A(b_0 + b_1 t + b_2 t^2 + \cdots + b_k t^k + \cdots)$$

Equating the coefficient of the equal power of t we obtain

$$
\begin{aligned}
b_1 &= Ab_0 \\
b_2 &= \frac{1}{2}Ab_1 = \frac{1}{2}A^2 b_0 \\
b_3 &= \frac{1}{3}Ab_2 = \frac{1}{6}A^3 b_0 \\
&\vdots \\
b_k &= \frac{1}{k!}A^k b_0
\end{aligned}
$$

By substituting $t = 0$ in $x(t)$ one can get $x(0) = b_0$. Thus the solution

$$
\begin{aligned}
x(t) &= (I + At + A^2\frac{t^2}{2!} + A^3\frac{t^3}{3!}\cdots + A^k\frac{t^k}{k!} + \cdots)x(0) \\
&= e^{At}x(0) \neq x(0)e^{At}
\end{aligned}
$$

Write $x(t) = \phi(t)x(0)$ where $\phi(t)$ is an $n \times n$ matrix and is the unique solution of

$$\dot{\phi}(t) = A\phi(t), \qquad \phi(0) = I$$

Note that $x(0) = \phi(0)x(0) = Ix(0)$ and $\dot{x}(t) = \dot{\phi}(t)x(0) = A\phi(t)x(0) = Ax(t)$

So, from the solution of the state equation

$$\phi(t) = e^{At} = \mathscr{L}^{-1}\{(sI - A)^{-1}\}$$
$$\phi^{-1}(t) = (e^{At})^{-1} = e^{-At} = \phi(-t)$$

Properties of state transition matrix for $\dot{x} = Ax$ where $\phi(t) = e^{At}$

1. $\phi(0) = e^{A0} = I$
2. $\phi(t) = e^{At} = (e^{-At})^{-1} = [\phi(-t)]^{-1}$
3. $\phi(t_1 + t_2) = e^{A(t_1+t_2)} = e^{At_1}e^{At_2} = \phi(t_1)\phi(t_2)$

Solution of non-homogeneous state equations

We shall begin by considering the matrix case

$$\dot{x}(t) = Ax(t) + Bu(t)$$
$$\dot{x}(t) - Ax(t) = Bu(t)$$

Left multiplication of both sides of this equation by e^{-At} gives

$$e^{-At}(\dot{x}(t) - Ax(t)) = e^{-At}Bu(t)$$
$$\frac{d}{dt}[e^{-At}x(t)] = e^{-At}Bu(t)$$

Integrating this equation from 0 to t gives

$$e^{-At}x(t) - x(0) = \int_0^t e^{-A\tau}Bu(\tau)d\tau$$
$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)}Bu(\tau)d\tau$$

The first term on the right-hand side is the response to the initial condition, and the second term is the response to the input $u(t)$.

Laplace transform approach for solution of non-homogeneous state equations

The solution of the non-homogeneous state equation can also be obtained by the Laplace transform approach

$$\dot{x}(t) - Ax(t) = Bu(t)$$

The Laplace transform of the above equation yields

$$\begin{aligned} sX(s) - x(0) &= Ax(s) + BU(s) \\ (sI - A)X(s) &= x(0) + BU(s) \\ X(s) &= (sI - A)^{-1}x(0) + (sI - A)^{-1}BU(s) \\ x(t) &= \mathscr{L}^{-1}\{(sI - A)^{-1}\}x(0) + \mathscr{L}^{-1}\{(sI - A)^{-1}BU(s)\} \end{aligned}$$

Thus far, we have assumed the initial time is zero. If, however, the initial time is given by t_0 instead of 0, then the solution to the state equation must be modified to

$$x(t) = e^{A(t-t_0)}x(t_0) + \int_{t_0}^{t} e^{A(t-\tau)}Bu(\tau)d\tau$$

Numerical Example

Obtain the time response of the following system

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

for u unit impulse and unit step.

To find state transition matrix $\phi(t) = \mathscr{L}^{-1}\{(sI - A)^{-1}\}$

$$(sI - A) = \begin{bmatrix} s & -1 \\ 2 & s+3 \end{bmatrix}$$

$$(sI - A)^{-1} = \frac{1}{s^2 + 3s + 2} \begin{bmatrix} s+3 & 1 \\ -2 & s \end{bmatrix}$$

$$= \begin{bmatrix} \frac{s+3}{(s+1)(s+2)} & \frac{1}{(s+1)(s+2)} \\ \frac{-2}{(s+1)(s+2)} & \frac{s}{(s+1)(s+2)} \end{bmatrix}$$

$$\mathscr{L}^{-1}\{(sI-A)^{-1}\} = \mathscr{L}^{-1}\begin{bmatrix} \frac{2}{(s+1)} + \frac{-1}{(s+2)} & \frac{1}{(s+1)} + \frac{-1}{(s+2)} \\ \frac{-2}{(s+1)} + \frac{2}{(s+2)} & \frac{-1}{(s+1)} + \frac{2}{(s+2)} \end{bmatrix}$$

The state transition matrix $\phi(t) = e^{At}$ was obtained earlier

$$\phi(t) = e^{At} = \begin{bmatrix} 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ -2e^{-t} + 2e^{-2t} & -e^{-t} + 2e^{-2t} \end{bmatrix}$$

The response to the unit impulse input is then obtained as

$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)}B\delta(\tau)d\tau.$$

As $\int_{0-\Delta}^{0+\Delta} f(t)\delta(t)dt = f(0)$ we have

$$x(t) = e^{At}(x(0) + B)$$

If the initial condition of state is zero $x(0) = 0$, then impulse response $x(t)$ is

$$\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} e^{-t} - e^{-2t} \\ -e^{-t} + 2e^{-2t} \end{bmatrix}$$

The response to the unit step input is then obtained as

$$x(t) = e^{At}x(0) + \int_0^t \begin{bmatrix} 2e^{-(t-\tau)} - e^{-2(t-\tau)} & e^{-(t-\tau)} - e^{-2(t-\tau)} \\ -2e^{-(t-\tau)} + 2e^{-2(t-\tau)} & -e^{-(t-\tau)} + 2e^{-2(t-\tau)} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} d\tau$$

$$\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ -2e^{-t} + 2e^{-2t} & -e^{-t} + 2e^{-2t} \end{bmatrix} \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix} + \begin{bmatrix} \frac{1}{2} - e^{-t} + \frac{1}{2}e^{-2t} \\ e^{-t} - e^{-2t} \end{bmatrix}$$

If the initial condition of state is zero, or $x(0) = 0$, then $x(t)$ can be simplified to

$$\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} \frac{1}{2} - e^{-t} + \frac{1}{2}e^{-2t} \\ e^{-t} - e^{-2t} \end{bmatrix}$$

Solution of Linear Time-Varying Systems

- Time-varying system can be solved from state space representation.
- This can be done by changing the transition matrix to $\phi(t, t_0)$ from $\phi(t)$.
- For time-varying systems, the transition matrix depends upon both t and t_0 and not the difference $t - t_0$. Thus, we cannot always set the initial time equal to zero.

Solution of time-varying autonomous state equation

For a Scalar differential equation (1) the solution can be given by (2)

$$\dot{x} = a(t)x(t) \tag{1}$$

$$x(t) = e^{\int_{t_0}^{t} a(\tau)d\tau} x(t_0) \tag{2}$$

The state transition function is given by

$$\phi(t, t_0) = e^{\int_{t_0}^{t} a(\tau)d\tau}$$

The same results, however, do not carry over to the vector matrix differential equation.

Consider the state equation

$$\dot{x} = A(t)x(t)$$

where $x(t)$ is n dimensional vector and $A(t)$ is $n \times n$ matrix whose elements are piecewise continuous function of t in the interval $t_0 \leq t \leq t_1$. The solution of this equation is given by

$$x = \phi(t, t_0)x(t_0)$$

where $\phi(t, t_0)$ is the $n \times n$ nonsingular matrix satisfying the following matrix differential equation

$$\dot{\phi}(t, t_0) = A(t)\phi(t, t_0), \qquad \phi(t_0, t_0) = I$$

The above fact that $x(t) = \phi(t, t_0)x(t_0)$ is the solution of the above equation can be verified easily as given below

$$x(t_0) = \phi(t_0, t_0)x(t_0) = Ix(t_0)$$

and

$$\begin{aligned}\dot{x}(t) &= \frac{d}{dt}[\phi(t, t_0)x(t_0)] \\ &= \dot{\phi}(t, t_0)x(t_0) \\ &= A(t)\phi(t, t_0)x(t_0) = A(t)x(t)\end{aligned}$$

It is important to note that the state transition matrix $\phi(t, t_0)$ is given by a matrix exponential if and only if $A(t)$ and $\int_{t_0}^{t} A(\tau)d\tau$ commute. That is

$$\phi(t, t_0) = e^{\left[\int_{t_0}^{t} A(\tau)d\tau\right]}$$

Note that if $A(t)$ is a constant matrix or diagonal matrix, $A(t)$ and $\int_{t_0}^{t} A(\tau)d\tau$ commute.

If $A(t)$ and $\int_{t_0}^{t} A(\tau)d\tau$ do not commute, then there is no simple way to compute the state transition matrix.

To compute $\phi(t, t_0)$ numerically, we may use the following series expansion for $\phi(t, t_0)$

$$\phi(t, t_0) = I + \int_{t_0}^{t} A(\tau) d\tau + \int_{t_0}^{t} A(\tau_1) \left[\int_{t_0}^{\tau_1} A(\tau_2) d\tau_2 \right] d\tau_1 + \cdots$$

It is called the Peano-Baker series.

Properties of the state transition matrix $\phi(t, t_0)$

[1] $\phi(t_2, t_1)\phi(t_1, t_0) = \phi(t_2, t_0)$
To prove this, note that

$$x(t_1) = \phi(t_1, t_0)x(t_0)$$
$$x(t_2) = \phi(t_2, t_0)x(t_0)$$

Also $x(t_2) = \phi(t_2, t_1)x(t_1)$ hence

$$\begin{aligned} x(t_2) &= \phi(t_2, t_1)\phi(t_1, t_0)x(t_0) \\ &= \phi(t_2, t_0)x(t_0) \end{aligned}$$

Thus

$$\phi(t_2, t_1)\phi(t_1, t_0) = \phi(t_2, t_0)$$

[2] $\phi(t_1, t_0) = \phi^{-1}(t_0, t_1)$
To prove this note that

$$\phi(t_1, t_0) = \phi^{-1}(t_2, t_1)\phi(t_2, t_0)$$

If we let $t_2 = t_0$ in this last equation, then

$$\phi(t_1, t_0) = \phi^{-1}(t_0, t_1)\phi(t_0, t_0) = \phi^{-1}(t_0, t_1)$$

Solution of linear time-varying state equations

Let us consider the system

$$\dot{x} = A(t)x(t) + B(t)u(t)$$

The elements of $A(t)$ and $B(t)$ are assumed to be piecewise continuous functions of t in the interval $t_0 \leq t \leq t_1$. In order to obtain the solution, let us put

$$x(t) = \phi(t, t_0)\xi(t)$$

where $\phi(t, t_0)$ is the unique matrix satisfying the following equation

$$\dot{\phi}(t, t_0) = A(t)\phi(t, t_0), \qquad \phi(t_0, t_0) = I$$

Then

$$\begin{aligned}\dot{x}(t) &= \frac{d}{dt}\left[\phi(t,t_0)\xi(t)\right] \\ &= \dot{\phi}(t,t_0)\xi(t) + \phi(t,t_0)\dot{\xi}(t) \\ &= A(t)\phi(t,t_0)\xi(t) + \phi(t,t_0)\dot{\xi}(t) \\ &= A(t)\phi(t,t_0)\xi(t) + B(t)u(t)\end{aligned}$$

Therefore $\phi(t,t_0)\dot{\xi}(t) = B(t)u(t)$ or $\dot{\xi}(t) = \phi^{-1}(t,t_0)B(t)u(t)$. Hence

$$\xi(t) = \xi(t_0) + \int_{t_0}^{t} \phi^{-1}(\tau,t_0)B(\tau)u(\tau)d\tau$$

Since $\xi(t) = \phi^{-1}(t,t_0)x(t) = x(t_0)$. The solution is obtained as

$$x(t) = \phi(t,t_0)x(t_0) + \phi(t,t_0)\int_{t_0}^{t} \phi^{-1}(\tau,t_0)B(\tau)u(\tau)d\tau$$

$$x(t) = \phi(t,t_0)x(t_0) + \int_{t_0}^{t} \phi(t,\tau)B(\tau)u(\tau)d\tau$$

Numerical Example

Compute $\phi(t,0)$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & t \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\int_0^t A(\tau)d\tau = \int_0^t \begin{bmatrix} 0 & 1 \\ 0 & \tau \end{bmatrix} d\tau$$
$$= \begin{bmatrix} 0 & t \\ 0 & \frac{t^2}{2} \end{bmatrix}$$

$$\int_0^t \begin{bmatrix} 0 & 1 \\ 0 & \tau \end{bmatrix} \left\{ \int_0^{\tau_1} \begin{bmatrix} 0 & 1 \\ 0 & \tau_2 \end{bmatrix} d\tau_2 \right\} d\tau_1 = \int_0^t \begin{bmatrix} 0 & 1 \\ 0 & \tau \end{bmatrix} \begin{bmatrix} 0 & \tau_1 \\ 0 & \frac{{\tau_1}^2}{2} \end{bmatrix} d\tau_1$$
$$= \begin{bmatrix} 0 & \frac{t^3}{6} \\ 0 & \frac{t^4}{8} \end{bmatrix}$$

State space representation of discrete-time systems

- The state space approach to the analysis of dynamic systems can be extended to the discrete-time case.
- The discrete form of the state space representation is quite analogous to the continuous form.

The most general state space representation of linear discrete-time system is

$$
\begin{aligned}
x(k+1) &= G(k)x(k) + H(k)u(k) \\
y(k) &= C(k)x(k) + D(k)u(k)
\end{aligned}
$$

where $x(k)$ is the state vector, $u(k)$ is the input vector and $y(k)$ is the output vector.

If the linear discrete-time systems are time-invariant then the above equations are modified to

$$
\begin{aligned}
x(k+1) &= Gx(k) + Hu(k) \\
y(k) &= Cx(k) + Du(k)
\end{aligned}
$$

Recursion procedure

In general, difference equations are easier to solve than differential equations because the former can be solved simply by means of a recursion procedure.

Consider the following difference equation -

$$x(k+1) + 0.2x(k) = 2u(k)$$

where $x(0) = 0$ and $u(k) = 1$ for k=0, 1, 2, ... The solution $x(1)$ can be found by recursion as

$$x(1) = -0.2x(0) + 2u(0) = 2$$

Similarly

$$\begin{aligned} x(2) &= -0.2x(1) + 2u(1) = 1.6 \\ x(3) &= -0.2x(2) + 2u(2) = 1.68 \\ x(4) &= -0.2x(3) + 2u(3) = 1.664 \end{aligned}$$

Note that in this method, $x(k+1)$ cannot be computed unless $x(k)$ is known. This procedure is quite simple and convenient for digital computations.

Solution of discrete-time state equations

The state space representation of a discrete LTI system is

$$\begin{aligned} x(k+1) &= Gx(k) + Hu(k) \\ y(k) &= Cx(k) + Du(k) \end{aligned}$$

The solution of this equation for any $k > 0$ may be obtained directly by recursion as follows.

$$\begin{aligned} x(1) &= Gx(0) + Hu(0) \\ x(2) &= Gx(1) + Hu(1) \\ &= G[Gx(0) + Hu(0)] + Hu(1) \\ &= G^2x(0) + GHu(0) + Hu(1) \\ x(3) &= Gx(2) + Hu(2) \\ &= G[G^2x(0) + GHu(0) + Hu(1)] + Hu(2) \\ &= G^3x(0) + G^2Hu(0) + GHu(1) + Hu(2) \end{aligned}$$

Proceeding similarly, we get

$$\begin{aligned} x(k) &= G^kx(0) + G^{k-1}Hu(0) + \ldots + GHu(k-2) + Hu(k-1) \\ &= G^kx(0) + \sum^{k-1} G^{k-j-1}Hu(j) \end{aligned}$$

Output Equation

- State Transition Matrix: $\Phi(k) = G^k$
- Properties of Φ(k): $\Phi(k+1) = G\Phi(k)$ and $\Phi(0) = I$.

Solution of States

$$x(k) = \Phi(k)x(0) + \sum_{j=0}^{k-1} \Phi(k-j-1)Hu(j) \tag{3}$$

Output Equation

$$\begin{aligned} y(k) &= C\Phi(k)x(0) + C\sum_{j=0}^{k-1} \Phi(k-j-1)Hu(j) + Du(k) \\ &= C\Phi(k)x(0) + C\sum_{j=0}^{k-1} \Phi(j)Hu(k-j-1) + Du(k) \end{aligned}$$

Z-Transform of the state equation

The state space equation of a discrete LTI system is

$$x(k+1) = Gx(k) + Hu(k)$$

Taking Z-transform on both sides of the above equation, we get

$$zX(z) - zX(0) = GX(z) + HU(z) \tag{4}$$

where $X(z) = Z[x(k)]$ and $U(z) = Z[u(k)]$. Then from (4),

$$(zI - G)X(z) = zX(0) + HU(z)$$

Premultiplying the above equation on both sides by $(zI - G)^{-1}$, we get

$$X(z) = (zI - G)^{-1}zX(0) + (zI - G)^{-1}HU(z) \tag{5}$$

Solution & Characteristic Equation

Solution using Z-Transform

Taking the inverse Z-transform of (5), we get

$$x(k) = Z^{-1}(zI - G)^{-1}zX(0) + Z^{-1}(zI - G)^{-1}HU(z)$$

So the state transition matrix

$$G^k = Z^{-1}(zI - G)^{-1}z$$

Characteristic Equation

The characteristic equation of the discrete-time system can be easily noted as

$$|zI - G| = 0$$

Example

Problem

Obtain the state transition matrix of the following discrete-time system

$$x(k+1) = Gx(k) + Hu(k)$$

where

$$G = \begin{bmatrix} 0 & 1 \\ -0.16 & -1 \end{bmatrix}, \quad H = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Then obtain x(k) when u(k)=1 for $k = 0, 1, 2, \ldots$. Assume that the initial condition is given by

$$x(0) = \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Example

Solution

We know that the state transition matrix $\Phi(k)$ is

$$\Phi(k) = G^k = Z^{-1}[(zI - G)^{-1}z]$$

Therefore, we first compute $(zI - G)^{-1}$.

$$\begin{aligned}(zI - G)^{-1} &= \begin{bmatrix} z & -1 \\ 0.16 & z+1 \end{bmatrix}^{-1} \\ &= \begin{bmatrix} \frac{z+1}{(z+0.2)(z+0.8)} & \frac{1}{(z+0.2)(z+0.8)} \\ \frac{-0.16}{(z+0.2)(z+0.8)} & \frac{z}{(z+0.2)(z+0.8)} \end{bmatrix}\end{aligned}$$

Thus, $\Phi(k)$ is obtained as

$$\begin{aligned}\Phi(k) = G^k &= Z^{-1}[(zI - G)^{-1}z] \\ &= Z^{-1}\begin{bmatrix} \frac{4}{3}\frac{z}{(z+0.2)} - \frac{1}{3}\frac{z}{(z+0.8)} & \frac{5}{3}\frac{z}{(z+0.2)} - \frac{5}{3}\frac{z}{(z+0.8)} \\ -\frac{0.8}{3}\frac{z}{(z+0.2)} + \frac{0.8}{3}\frac{z}{(z+0.8)} & -\frac{1}{r}3\frac{z}{(z+0.2)} + \frac{4}{3}\frac{z}{(z+0.8)} \end{bmatrix} \\ &= \begin{bmatrix} \frac{4}{3}(-0.2)^k - \frac{1}{3}(-0.8)^k & \frac{5}{3}(-0.2)^k - \frac{5}{3}(-0.8)^k \\ -\frac{0.8}{3}(-0.2)^k + \frac{0.8}{3}(-0.8)^k & -\frac{1}{3}(-0.2)^k + \frac{4}{3}(-0.8)^k \end{bmatrix}\end{aligned}$$

Example

Solution Contd...

Next, we compute $x(k)$. The Z-transform of $x(k)$ is given by

$$\begin{aligned} Z[x(k)] = X(z) &= (zI - G)^{-1} zX(0) + (zI - G)^{-1} HU(z) \\ &= (zI - G)^{-1}[zX(0) + HU(z)] \end{aligned}$$

Since $U(z) = \frac{z}{z-1}$, we obtain

$$zX(0) + HU(z) = \begin{bmatrix} z \\ -z \end{bmatrix} + \begin{bmatrix} \frac{z}{z-1} \\ \frac{z}{z-1} \end{bmatrix} = \begin{bmatrix} \frac{z^2}{z-1} \\ \frac{-z^2+2z}{z-1} \end{bmatrix}$$

Hence

$$X(z) = (zI - G)^{-1}[zX(0) + HU(z)] \qquad = \begin{bmatrix} \frac{(z^2+2)z}{(z+0.2)(z+0.8)(z-1)} \\ \frac{(-z+1.84)z^2}{(z+0.2)(z+0.8)(z-1)} \end{bmatrix}$$

Thus,

$$x(k) = Z^{-1}[X(z)] = \begin{bmatrix} -\frac{17}{6}(-0.2)^k + \frac{22}{9}(-0.8)^k + \frac{25}{18} \\ \frac{3.4}{6}(-0.2)^k - \frac{17}{6}(-0.8)^k + \frac{7}{18} \end{bmatrix}$$

THANK YOU